

EAZYAI INTELLIGENCE

ENTERPRISE WHITEPAPER

Autonomous Talent Intelligence & Multimodal AI Interviewing

Complete technical architecture, zero-trust security specifications, distributed microservices catalog, proctored anti-fraud engine, and glassmorphic design system implementation.

FRAMEWORK CORE

TanStack Start + React 19

RUNTIME & CLOUD

Nitro ESM & Google Cloud Run

SECURITY MODEL

IAM Zero-Trust + HttpOnly Cookie

CLASSIFICATION

Public Technical Specification

EXECUTIVE BRIEF

1. Platform Overview & Problem Statement

Modern enterprise recruitment faces severe bottlenecks: high manual screening overhead, vulnerability to synthetic resumes, impersonation during remote assessments, and lack of calibrated technical rubrics. EazyAI solves this through an end-to-end autonomous recruiting mesh combining semantic vector search, conversational AI agents, multimodal evaluation, and real-time biometric anti-fraud proctoring.

1. Vector RAG Matching

Automated resume parsing and semantic cosine similarity scoring across candidate archives with explicit gap analysis and seniority deduction.

2. Multimodal AI Interviews

Autonomous spoken technical interviews conducted via Dialogflow CX, featuring real-time speech-to-text, acoustic waveform inspection, and domain rubric scoring.

3. Anti-Fraud Proctoring

Synthetic voice deepfake detection, computer vision webcam face verification, and proctored movement anomaly tracking combined into a unified Trust Gauge.

INDEX

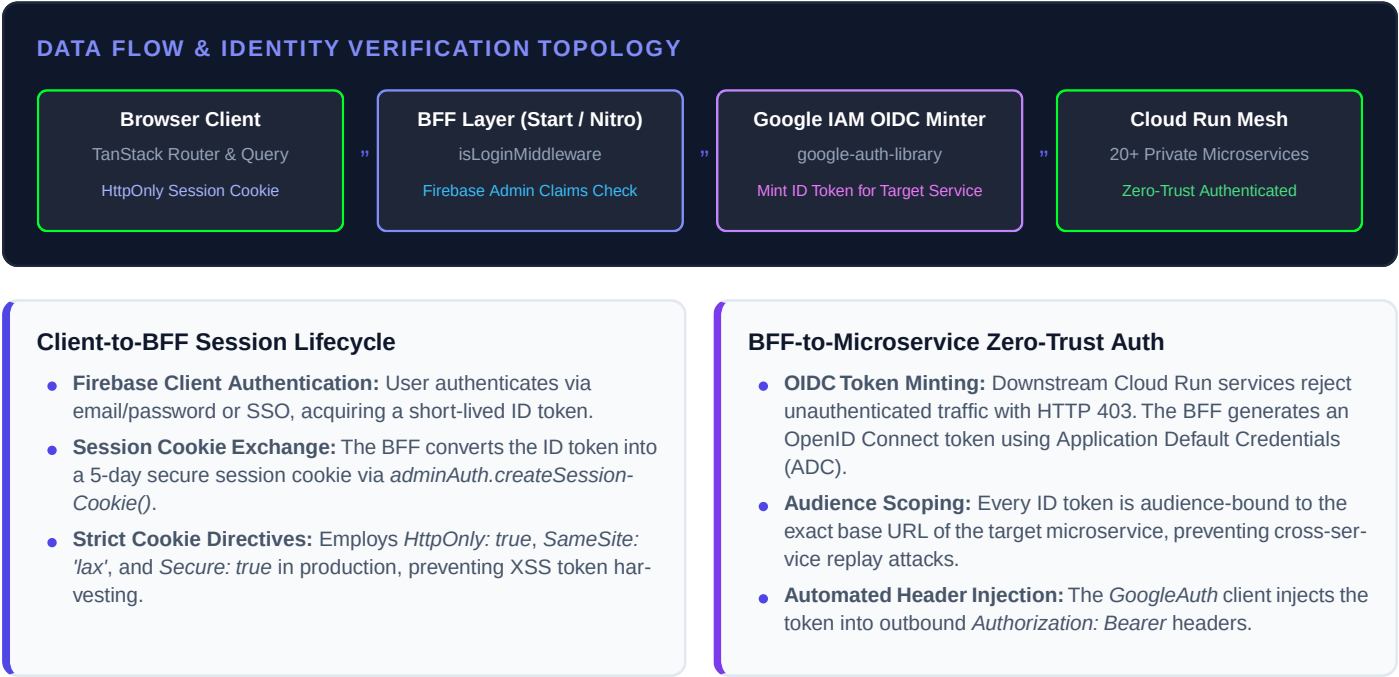
Document Structure

SECTION	TOPIC	CORE ARCHITECTURAL FOCUS
Section 2	System Architecture & Security Model	BFF Pattern, Nitro, Google Cloud IAM & OIDC
Section 3	Microservices Ecosystem & API Registry	20+ Cloud Run Endpoints, Dialogflow CX & OpenRouter
Section 4	Candidate Ingestion, RAG Discovery & Questions	Storage Explorer, Vector Corpus & Sandbox Simulator
Section 5	Multimodal Intelligence & Anti-Fraud Engine	Deepfake Voice Analysis, Gaze Proctoring, Trust Gauge
Section 6	Evaluator Workbench, Email Sync & Admin	Scratchpad, PDF Dossiers, Inbound Mail & RBAC
Section 7	Glassmorphism Design & Engineering Guide	OKLCH Palette, Tailwind v4, Vitest & Docker Deployment

ARCHITECTURE

2. Backend-For-Frontend (BFF) & Zero-Trust Security

EazyAI adheres to strict separation of concerns. Browser clients never communicate directly with downstream microservices. The TanStack Start server layer executes within the Nitro engine, serving as the sole trusted orchestrator that authenticates requests, enforces role-based access, and securely proxies payloads to Google Cloud Run.



INTEGRATION MESH

3. Microservices Topology & External AI Gateways

All downstream communication is bound through the centralized registry in `src/lib/api-path.ts`. The platform orchestrates 20+ specialized Google Cloud Run containers deployed across *us-central1* and *europa-west1*.

DOMAIN / SERVICE KEY	REGION	METHOD	FUNCTIONAL RESPONSIBILITY
BUCKET_LIST_API	us-central1	GET	Inspects GCS storage prefixes for resume PDFs and media assets.
RAG_SEARCH_API	us-central1	POST	Executes semantic vector searches over parsed resumes against job criteria.
TRIGGER_INDEX	us-central1	POST	Triggers asynchronous re-indexing into Corpus 1939767209815441408.
JOB_DETAILS / ADD / EDIT	us-central1	CRUD	Requisition lifecycle management, date windows, multi-location arrays.
ADD_MULTIPLE_CANDIDATES	us-central1	POST	High-throughput batch resume ingestion using multipart/form-data.
QUESTION_ADD_AI	us-central1	POST	AI generator synthesizing technical and behavioral interview questions.
SPEECH_TO_TEXT	us-central1	POST	Transcribes spoken candidate audio answers with timing markers.
VOICE_FRAUD_DETECTION	us-central1	GET	Detects synthetic voice cloning, audio artifacts, and deepfake speech.
FACE_DETECTION	us-central1	POST	Compares webcam capture against resume photo to prevent impersonation.
MOVEMENT_OUTCOME	us-central1	GET	Monitors candidate tab switching, gaze deviations, and head orientations.
INTERVIEW_VIDEO_API	us-central1	GET	Streams candidate interview video recordings directly from Google Cloud Storage.

Google Cloud Dialogflow CX Integration

EazyAI integrates natively with the Dialogflow CX runtime via `@google-cloud/dialogflow-cx`. The conversational agent manages stateful interview dialog, transitions through technical assessment nodes, and logs real-time token metrics (Prompt, Candidate, Audio, and Text tokens) visible in the session timeline.

OpenRouter LLM Synthesis Gateway

The platform leverages `@openrouter/sdk` to invoke state-of-the-art models such as *nvidia/nemotron-3-super-120b-a12b:free*. Recruiters input minimal parameters (Title, Experience, Skills) to generate calibrated job descriptions and rubric standards.

Downstream Microservice Registration Principle: All new services must be registered in `API_PATH`, have typed interfaces in `src/lib/types.ts`, and be wrapped in a `createServerFn` with `isLoginMiddleware` to ensure uniform error handling, telemetry, and zero-trust authentication.

INGESTION & DISCOVERY

4. Resume Ingestion, Vector Archive & RAG Matching

EazyAI combines cloud-native object storage with high-dimensional vector embeddings to automate candidate sourcing, eliminating reliance on keyword filters in favor of deep semantic qualifications matching.

Archive Bank & Cloud Storage Explorer

The Archive Bank (*/dashboard/import*) provides a virtual file system inspecting Google Cloud Storage. Recruiters can view folder hierarchies, review MIME metadata, download signed URLs, and select specific document subsets to trigger asynchronous vector re-indexing into Corpus **1939767209815441408**.

Batch High-Throughput CV Ingestion

The Batch Ingestion modal (*add-multiple-candidates-dialog.tsx*) supports drag-and-drop batch upload of dozens of resumes concurrently. The BFF streams files via *multipart/form-data* to the ingestion microservice, parsing candidate names, emails, and contact records with zero UI blocking.

SEMANTIC SEARCH

Vector RAG Discovery Architecture

The AI Discovery engine (*/dashboard/discover*) allows recruiters to articulate nuanced hiring criteria through a structured form validated via `profileSearchSchema`:

EVALUATION DIMENSION	PROCESSING & VECTOR MECHANICS	RECRUITER ACTIONABLE OUTPUT
Match Score (0-100%)	Cosine similarity between requisition embedding and resume text chunk vectors.	Color-coded match badge for instant top-tier candidate ranking.
Seniority Inference	Heuristic deduction across total experience, team leadership indicators, and roles.	Categorized as Junior, Mid, Senior, or Lead architect level.
Matched Criteria	Vector projection highlights matching skills, libraries, and certifications.	List of explicitly satisfied job requirements with source references.
Missing Information	Negative constraint analysis identifying requisition requirements omitted in CV.	Flags skill gaps for probing during live automated interviews.

QUESTION SANDBOX

AI Question Bank & Candidate Experience Simulator

Calibrated Question Bank

Recruiters can manually curate or synthesize interview questions per requisition. The AI Question Generator (*QUESTION_ADD_AI*) constructs questions balanced across domain technical knowledge, system architecture, and behavioral leadership rubrics.

Interactive Candidate Simulator

The Question Simulator (*question-simulator-dialog.tsx*) allows hiring teams to experience the interview from the applicant perspective. It emulates question pacing, speech input, countdown timers, and simulated audio response transcription before questions are deployed live.

MULTIMODAL AI

5. Automated Interview Evaluation & Biometric Anti-Fraud

Remote recruitment is increasingly vulnerable to AI-assisted cheating, proxy test takers, and synthetic speech cloning. EazyAI combines acoustic waveform forensics, biometric face verification, and proctoring telemetry into an impregnable trust layer.

Face Anti-Impersonation

Compares webcam captures taken during the interview against candidate resume photos using neural face embeddings (- *FACE_DETECTION*). Prevents proxy candidate substitutions.

Voice Deepfake Detection

The Audio Analysis engine (- *VOICE_FRAUD_DETECTION*) inspects acoustic spectral flux, cloning jitter, and synthetic artifacts, classifying recordings as **Human**, **Suspicious**, or **AI-Generated**.

Movement & Gaze Proctoring

Telemetry tracking logs browser tab switches, background application focus changes, out-of-frame head deviations, and erratic gaze shifts during active questioning (*MOVEMENT_OUTCOME*).

AUDIO & VIDEO

Acoustic Waveform Analysis & Video Stream Review

HTML5 Web Audio API Waveform Player

The acoustic player (*audio-waveform-player.tsx*) creates an in-browser **AudioContext** with real-time frequency-domain and time-domain visualizers. Evaluators can scrub audio, modulate playback rates (0.75x to 2.0x), and inspect acoustic anomalies during candidate responses.

Candidate Video Streaming & Outcome Preview

The Video Outcome suite (*video-outcome.tsx*) streams interview video recordings stored in Google Cloud Storage via authenticated signed URLs. Displays duration, file size, timestamps, and playback controls directly aligned with candidate transcripts.

TRUST SCORING

Composite Integrity Trust Gauge Architecture

BIOMETRIC TELEMETRY	WEIGHT CONTRIBUTION	VERIFICATION THRESHOLD	PENALTY TRIGGER
Face Verification	35% Total Weight	Cosine Match > 0.82	Impersonation flag, unknown attendee
Acoustic Authenticity	35% Total Weight	Human Confidence > 85%	Synthetic speech / Voice cloning detected
Movement & Tab Switches	30% Total Weight	< 3 Focus Changes	Continuous gaze off-screen, tab switching

Session Timeline Transparency: The Session Timeline (*session-timeline.tsx*) reconstructs the complete Dialogflow CX conversational journey. Hiring teams can review every turn, tool invocation arguments, candidate thought signatures, and exact token counts.

EVALUATOR COPILOT

6. Evaluator Workbench Copilot & Executive Dossiers

EazyAI provides a client-side copilot environment empowering recruitment teams to conduct rigorous, synchronized evaluations with real-time composite scoring and client-rendered executive PDF generation.

Evaluator Scratchpad Copilot

The Scratchpad (*evaluator-scratchpad.tsx*) automatically persists evaluator notes in **localStorage** keyed by candidate and job ID. Includes weighted scoring sliders (Technical Competency, Communication, Cultural Fit) and one-click verdict templates ("Strong Hire", "Solid Contender", "Reject - Technical Gap", "Reject - Integrity Flag").

Client-Side Executive Dossier Generation

Powered by **@react-pdf/renderer**, the platform dynamically renders multi-page PDF candidate evaluation dossiers without requiring backend headless browsers. Dossiers encapsulate candidate demographics, Trust Gauge biometric scores, question breakdowns, and evaluator signatures.

ENTERPRISE GOVERNANCE

Inbound Email Sync, RBAC & Site Configuration

Inbound Email Sync

Synchronizes with recruitment mailboxes (*EMAIL_SYNC*), parsing applicant resumes, extracting candidate email bodies, and auto-associating candidates with active job IDs.

Role-Based Access Control

Admin Management (*/dashboard/admin-user*) supports granular roles: Super Admin, Recruiter, Interviewer, and Viewer. Includes account restriction toggles and access audit logs.

Global Interview Config

Configuration console (*/dashboard/config*) controls interview duration quotas, candidate invitation link expiration windows (e.g. 48 hours), and question count limits.

PRODUCTIVITY

Command Palette (K) & Guided Onboarding Tour

PRODUCTIVITY FEATURE	COMPONENT IMPLEMENTATION	USER WORKFLOW ADVANTAGE
Global Search (K / Ctrl+K)	global-search-dialog.tsx	Instant keyboard navigation across jobs, candidate files, routes, and admin tools.
Interactive Tour	onboarding-tour-dialog.tsx	Interactive multi-step modal tour welcoming new recruiters, stored in local-Storage.
India Requisition Map	facility-map-dialog.tsx	Interactive SVG visualization of job distribution across all 28 states and union territories.

DESIGN SYSTEM

7. Frosted Glassmorphism Design System (OKLCH)

EazyAI is designed to deliver a visually captivating, depth-rich user experience. Built with **Tailwind CSS v4**, the interface utilizes the **OKLCH** color space for superior perceptual contrast and vibrancy when interacting with semi-transparent frosted glass layers.

DESIGN TOKEN CLASS	TAILWIND V4 STYLING RULE	OKLCH LIGHT / DARK	UI PLACEMENT ROLE
glass-header	bg-background/50, backdrop-blur-2xl	Light: 50% / Dark: 70%	Sticky top navigation bar
glass-sidebar	bg-sidebar/40, backdrop-blur-2xl	Light: 40% / Dark: 40%	Primary app navigation panel
glass-card	bg-card/30, backdrop-blur-lg, shadow-lg	Light: 30% / Dark: 30%	Metric widgets, data tables, dialogs
ambient-pulse	15s breathing pulse keyframe	Indigo / Violet glow	Dual corner ambient background

TYPOGRAPHY

Typography Hierarchy

Outfit Variable

Used for page heroes, section headings, and stat values. Provides a modern, geometric SaaS brand identity with tight tracking.

Geist Sans Variable

The primary sans-serif font applied to body copy, form labels, data table cells, and button interfaces.

Geist Mono Variable

Monospace font applied to timestamps, session IDs, token metrics, and technical API payloads.

ENGINEERING & DEVOPS

Development Toolchain, Quality Assurance & Deployment

Development Workflows & Testing

- Package Management:** Enforces *pnpm* with pinned dependencies in *pnpm-lock.yaml*.
- Unit Testing with Vitest:** *pnpm test* executes component and utility suites with mocked browser media APIs.
- Formatting & Linting:** Standardized via TanStack ESLint configuration and Prettier (*pnpm check*).

Production Container & Cloud Deployment

- Multi-Stage Dockerfile:** Stage 1 builds the Nitro ESM bundle; Stage 2 prunes devDependencies; Stage 3 runs lean Node 20 slim image on port 8080.
- Firebase App Hosting:** Configured via *apphosting.yaml* with Google Cloud Secret Manager integration and dynamic autoscaling (0–10 instances, concurrency 80).